

ASSIGNMENT 1 – N8N CASE STUDIES

Enterprise Workflow Automation Analysis

Objective: Analyze real-world enterprise implementations of n8n, focusing on business challenges, workflow design, technical integration, and operational impact.

Case Study 1: Delivery Hero

DEVOPS & INTERNAL TOOLING AUTOMATION

1. Problem & Use Case

- Managing custom operational scripts, database updates, and multi-tool approval workflows across internal teams was creating heavy engineering bottlenecks and slow resolution times.

2. Implementation Rationale

- Deployed within core IT & DevOps infrastructure. n8n was selected as a self-hosted, secure automation engine to connect internal microservices and third-party tools without custom boilerplate code.

3. Workflow & Key Nodes

- **Webhook / Slack Node:** Listens for admin/employee trigger requests directly in Slack channels.
- **Switch / IF Node:** Evaluates user permission levels and payload parameters.
- **PostgreSQL / HTTP Request Node:** Interacts safely with internal databases and microservice REST APIs to execute approved actions.
- **Slack Node (Response):** Posts real-time execution updates back to the requesting user.

4. Real-World Business Impact

- Reduced manual DevOps ticket execution times from hours to seconds and eliminated maintenance overhead for custom internal tools.

Case Study 2: JumpStory

MARKETING OPERATIONS & CUSTOMER ONBOARDING

1. Problem & Use Case

- Onboarding new trial users, enriching customer profiles, and triggering personalized email sequences across disparate tools (Stripe, CRM, SendGrid) was fragmented and prone to manual tracking errors.

2. Implementation Rationale

- Implemented across Marketing & Growth engines to manage customer lifecycle events. Chosen for its flexible conditional branching based on real-time user behavior.

3. Workflow & Key Nodes

- **Webhook Node:** Captures real-time sign-up and payment events from Stripe or web forms.
- **Code / Set Node:** Cleans, normalizes, and structures incoming customer payload data.
- **Clearbit / HTTP Request Node:** Enriches user profiles with domain and company insights.
- **SendGrid Node:** Triggers targeted drip campaigns based on user segment and onboarding stage.

4. Real-World Business Impact

- Automated 100% of the customer onboarding journey, resulting in higher trial-to-paid conversion rates without expanding marketing ops headcount.

Case Study 3: Global E-Commerce Operations

AI-POWERED CUSTOMER SUPPORT & TICKETING

1. Problem & Use Case

- High volume of repetitive customer inquiries (Order Status, Returns, Refund Requests) overwhelming human support staff and delaying response times.

2. Implementation Rationale

- Integrated directly into the Customer Support Helpdesk pipeline. Selected due to native support for LangChain and OpenAI nodes alongside standard database/CRM connectors.

3. Workflow & Key Nodes

- **Zendesk Node:** Listens for incoming support tickets.
- **OpenAI / AI Agent Node:** Parses sentiment, categorizes customer intent, and drafts contextual responses.
- **Shopify / ERP HTTP Node:** Queries internal databases for live shipment and order status details.
- **Switch Node:** Auto-responds if confidence is high, or routes ticket to human agents with pre-drafted context.

4. Real-World Business Impact

- Reduced overall ticket resolution times by over 60% and allowed human agents to focus strictly on complex edge-case inquiries.

Enterprise Implementation Summary

Enterprise	Primary Use Case	Key Integrations / Nodes	Main Business Value
Delivery Hero	DevOps & Internal Tooling	Slack, Postgres, HTTP Request, Switch	Eliminated dev bottlenecks & automated internal approvals
JumpStory	Customer Onboarding & Growth	Stripe, Webhooks, SendGrid, Clearbit	100% automated user lifecycle & improved conversion
E-Commerce Retailer	AI Support Ticketing	Zendesk, OpenAI, Shopify/ERP, Switch	60% faster resolution times & auto-classified tickets